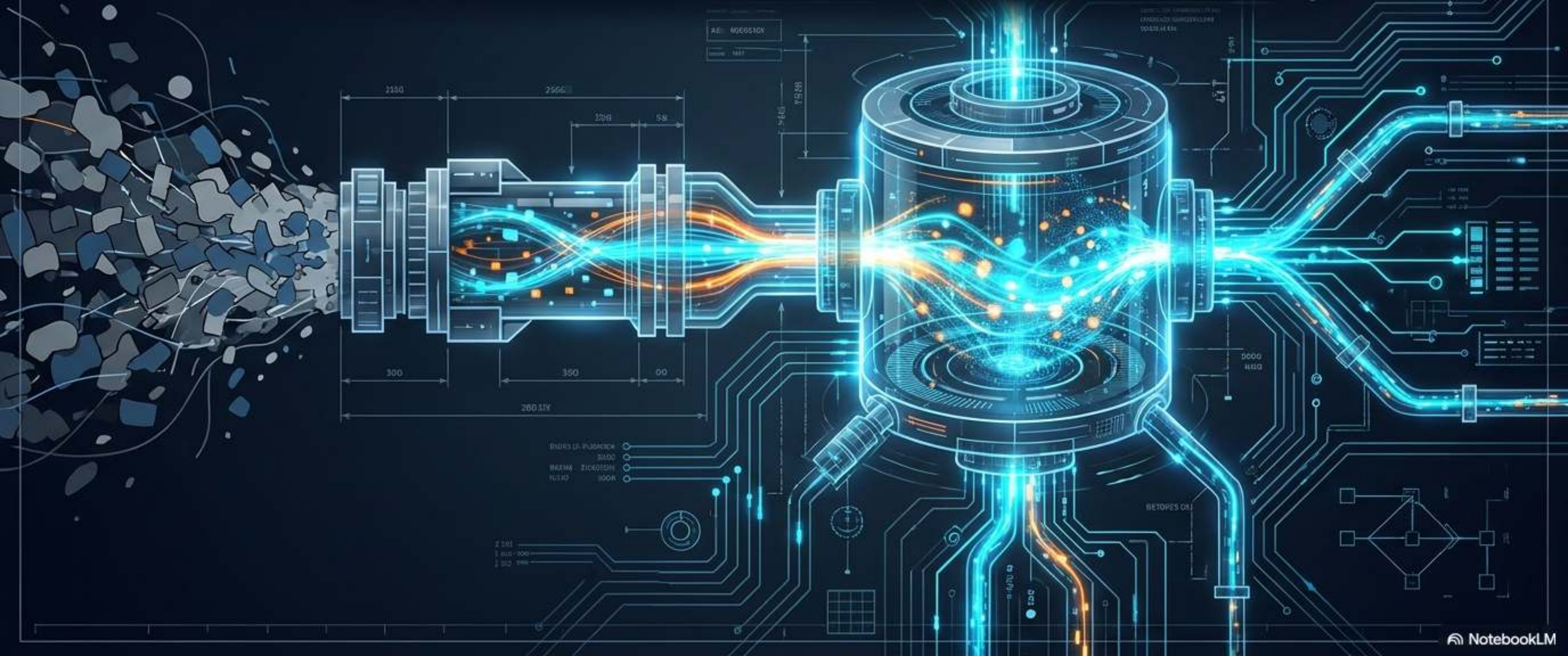
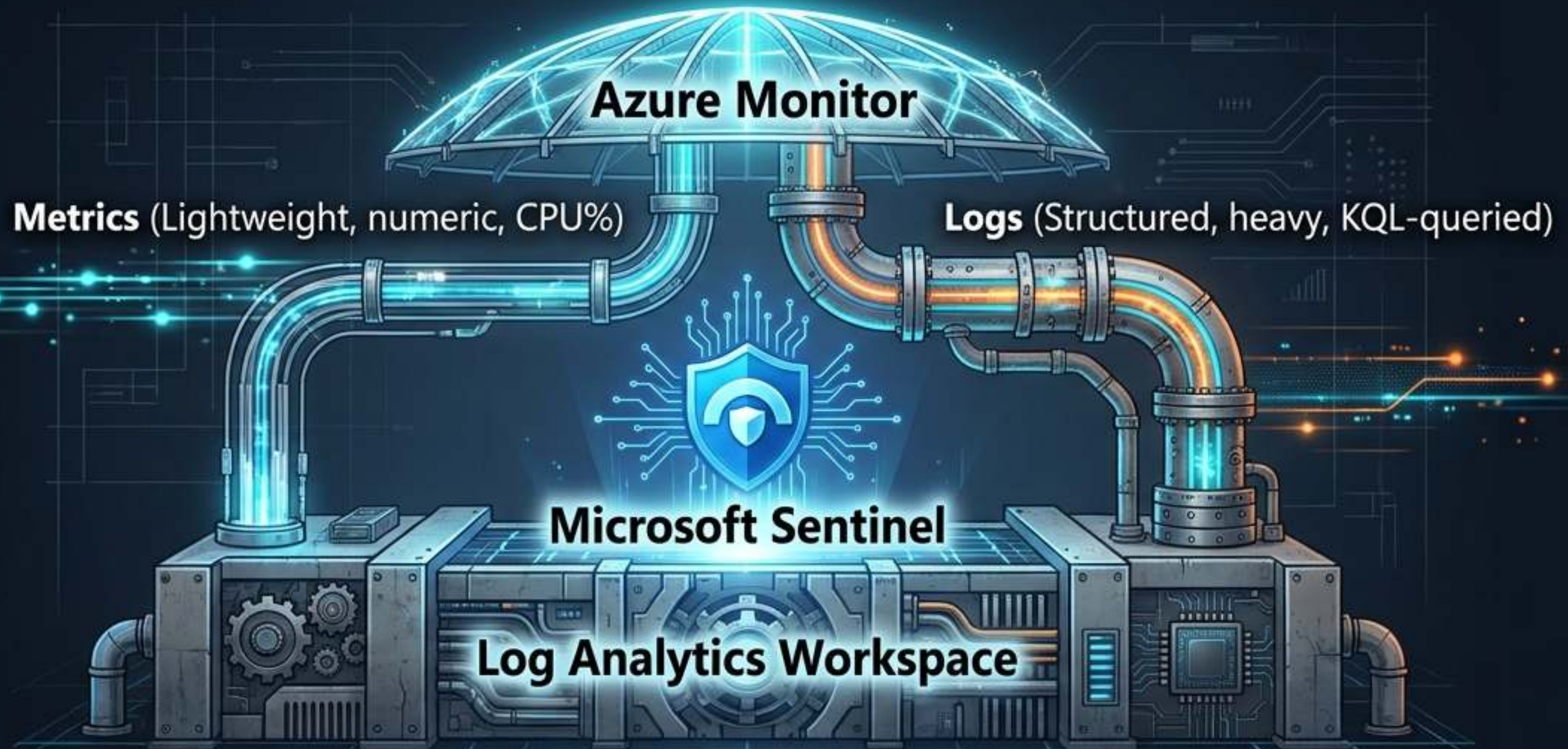


THE TELEMETRY PIPELINE

Azure Monitor & Data Collection Rules (AZ-500 Exam Prep)



Microsoft Sentinel rides on the Log Analytics foundation



Synthesis Note: Sentinel is an analytical engine, not a standalone storage resource. Enabling it applies SIEM/SOAR capabilities directly against the existing Log Analytics workspace.

The legacy paradigm functions as a rigid, expensive firehose

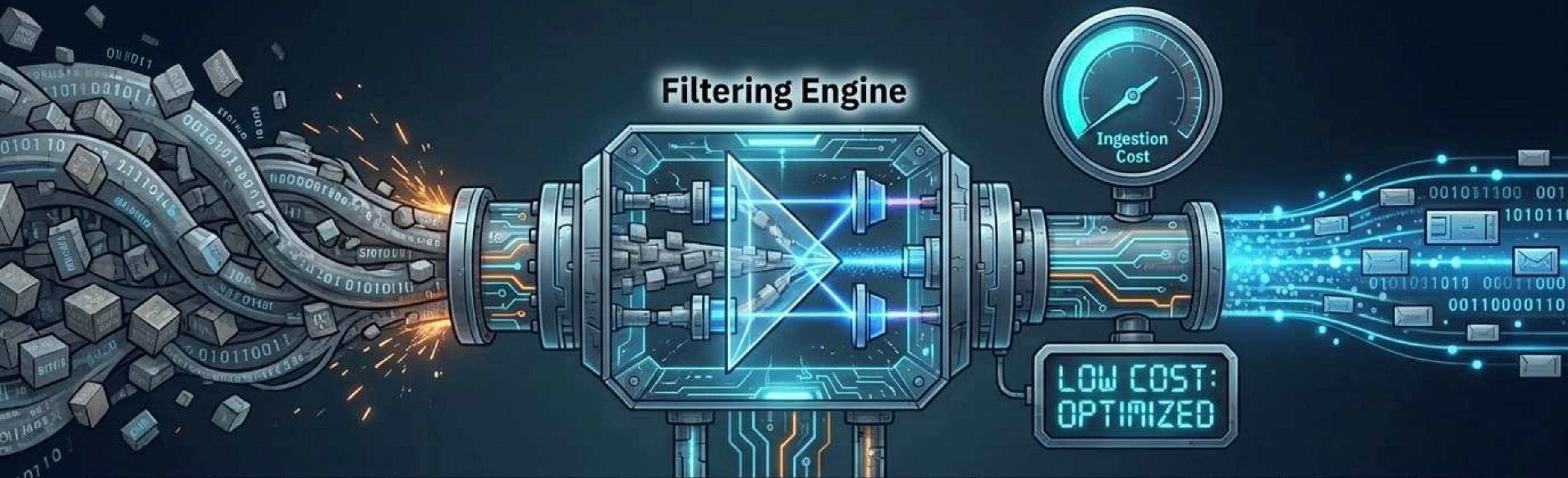


Mechanism: Diagnostic Settings

Behavior: Sends resource logs directly to fixed destinations.

The Problem: It is an **all-or-nothing pipeline**. You cannot filter or reshape the data before it lands, leading to **massive ingestion costs** for low-value logs.

The modern pipeline acts as a smart, transformable filter



Mechanism: Data Collection Rules (DCRs) + Data Collection Endpoints (DCEs)

Behavior: Granular, configurable, and filterable.

The Solution: Allows organizations to reshape, filter, and route telemetry before it hits the database, decisively controlling ingestion costs.

Anatomy of a Data Collection Rule



1. Sources

Where does it come from?

Windows Events,
Syslog, Perf Counters

2. Streams

How is it formatted?

Schema mapping

3. Transformations

How is it reshaped?

The in-flight
KQL engine

4. Destinations

Where does it go?

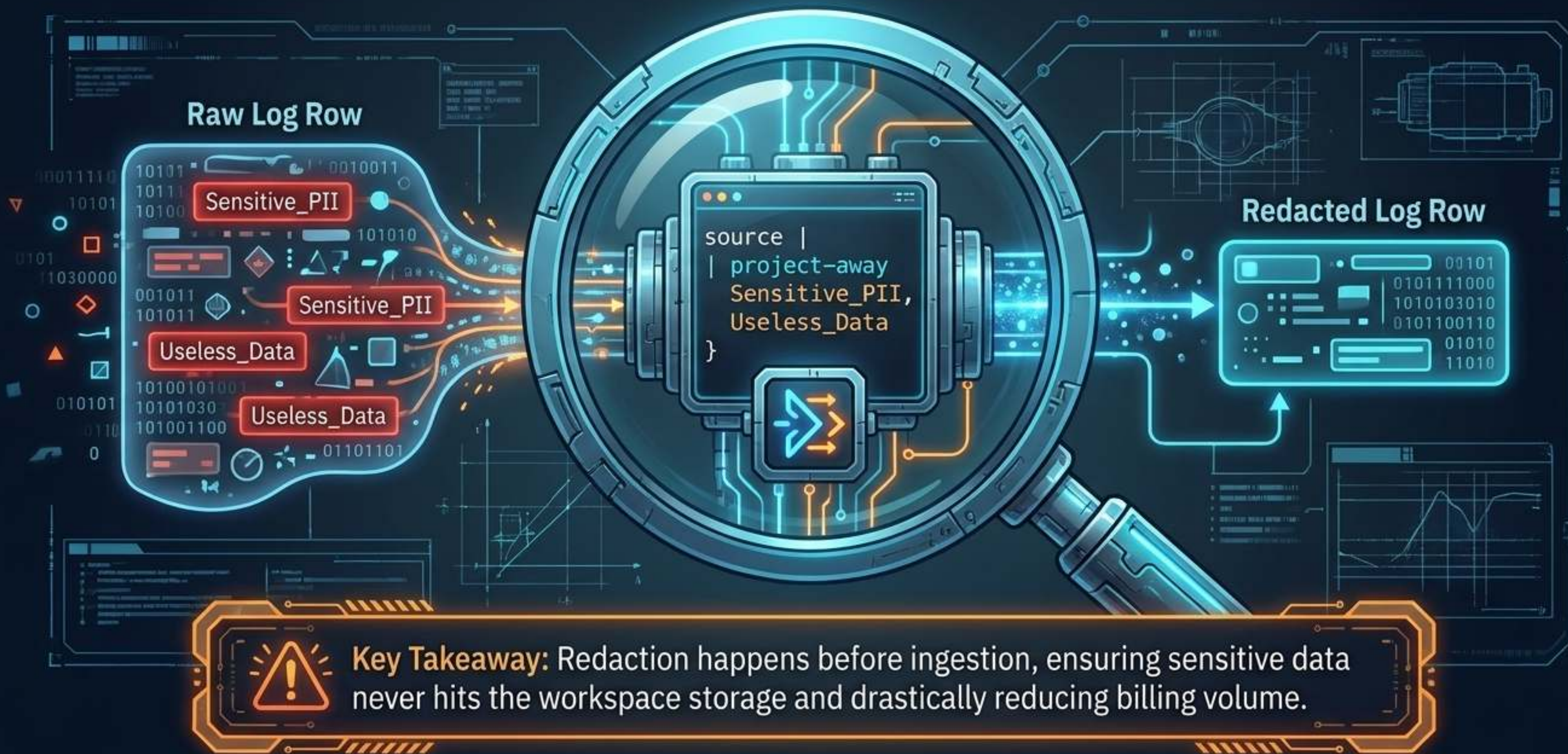
Specific Log
Analytics tables

5. Associations / DCR-A

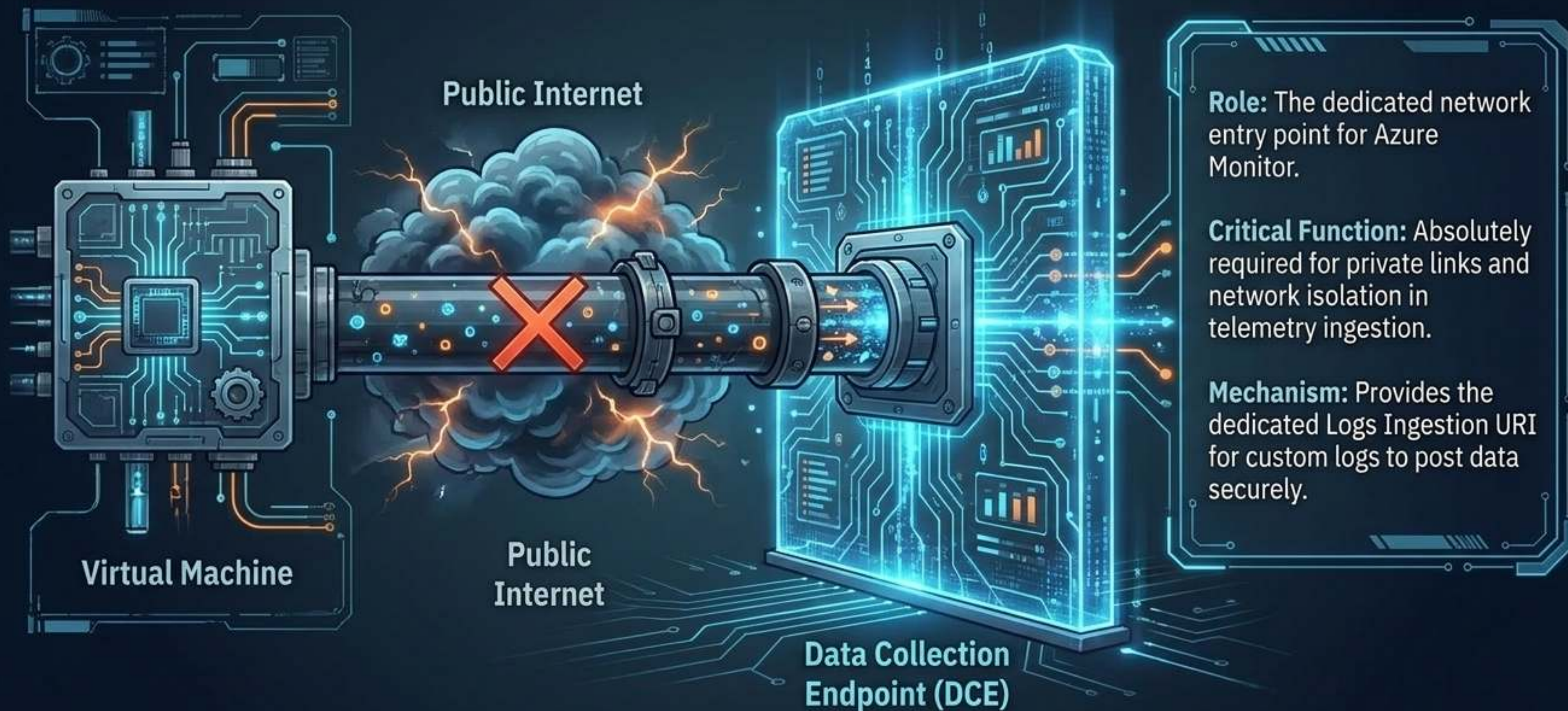
Who does it apply to?

The glue linking the rule
to VMs or Arc-enabled
servers

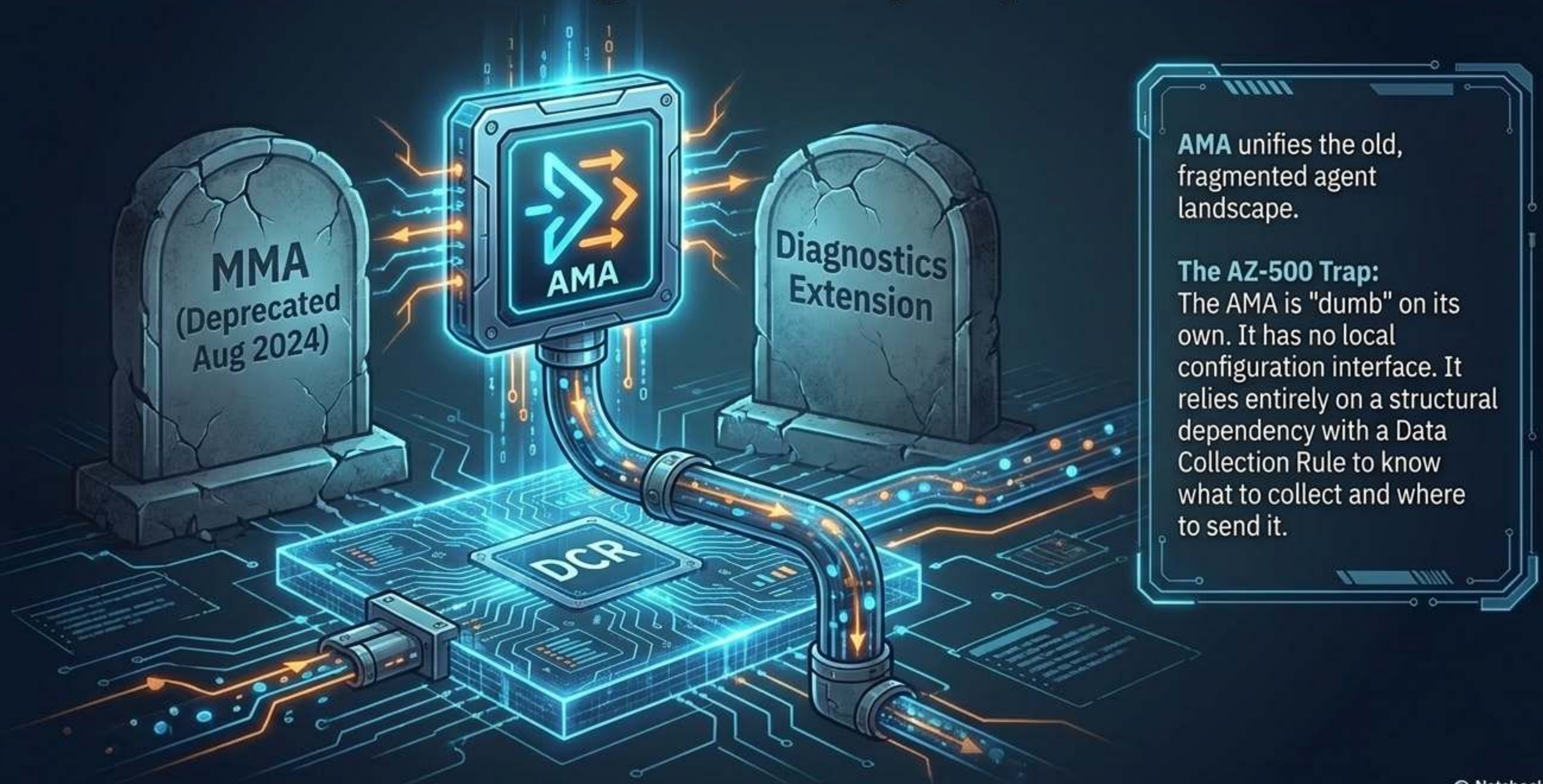
KQL transformations redact and refine data in flight



Data Collection Endpoints secure the network perimeter



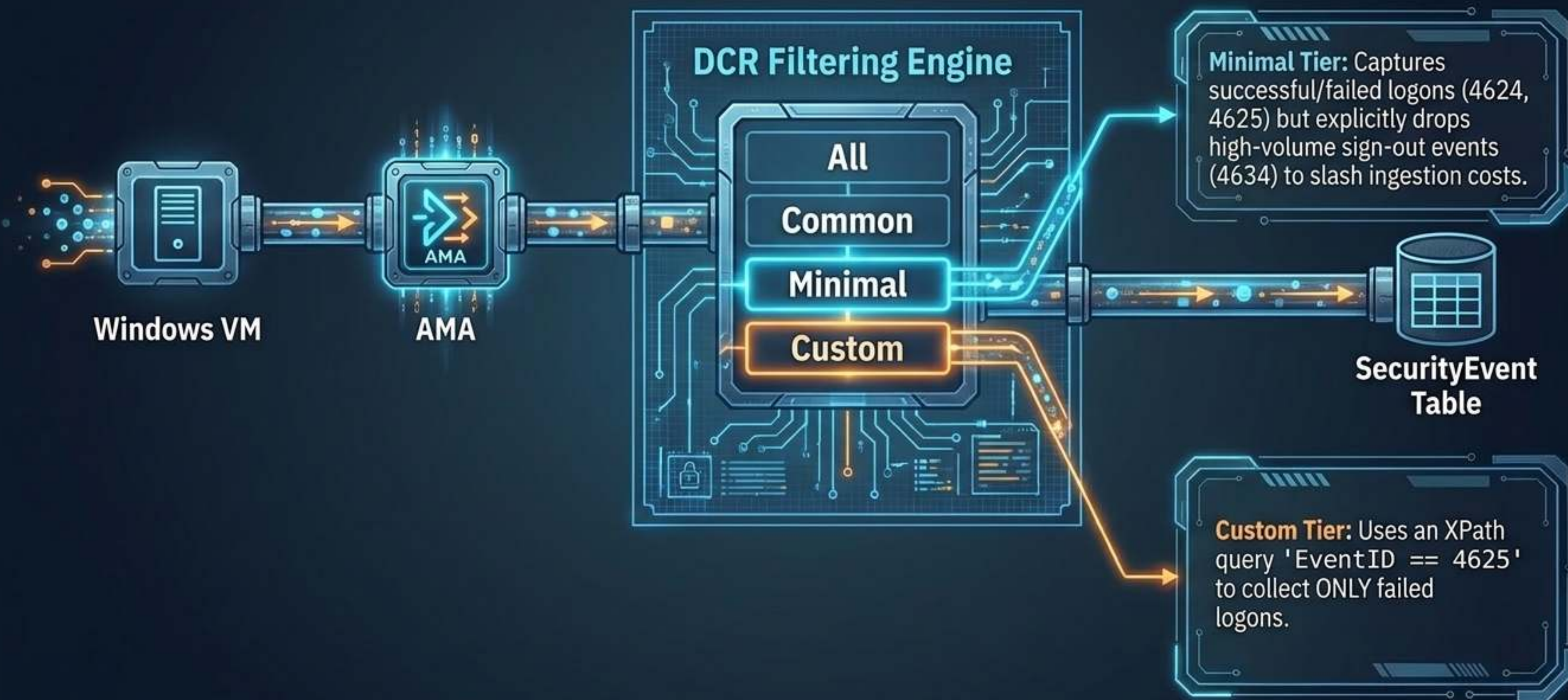
The Azure Monitor Agent strictly depends on the DCR



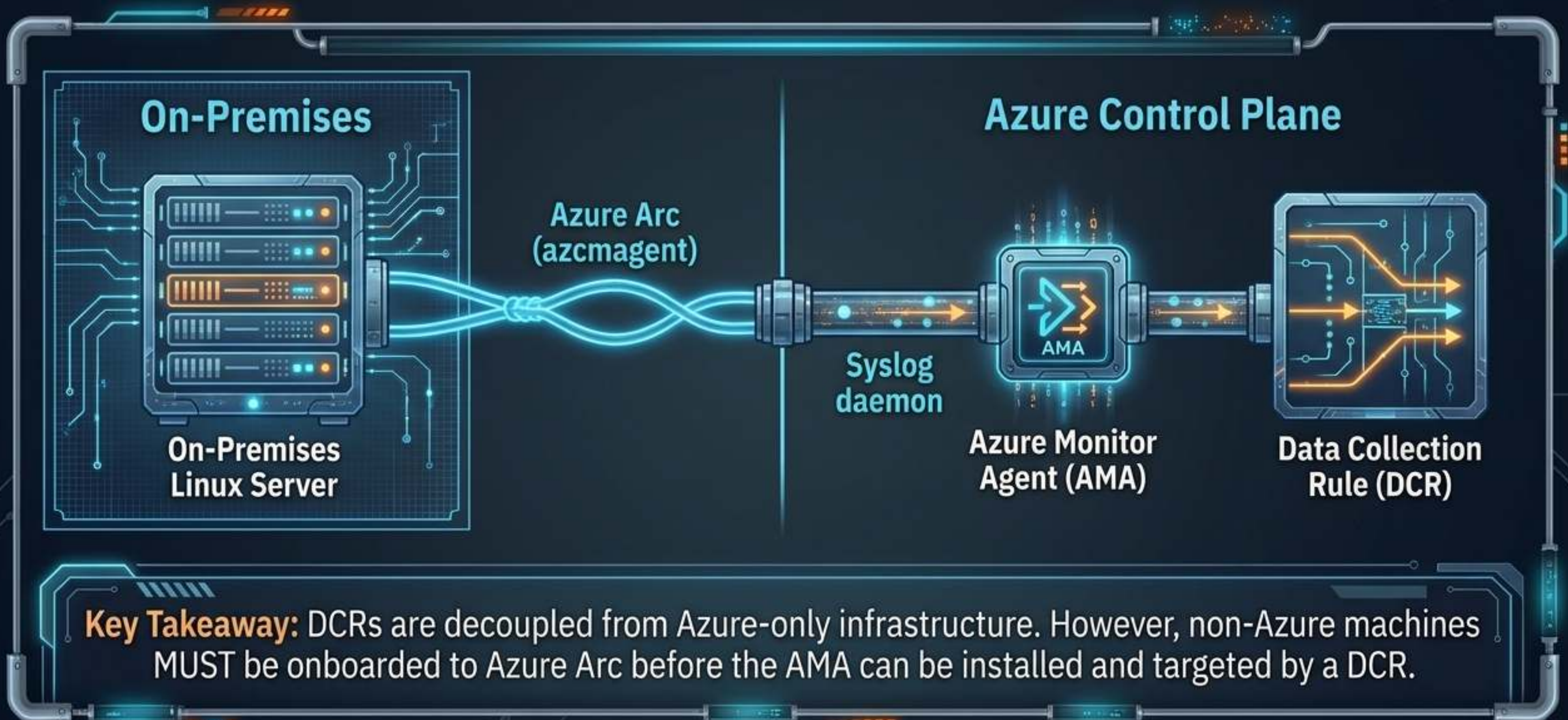
The telemetry pipeline component matrix

Component	What It Defines	Where It's Configured	Strict Dependencies
Diagnostic Settings (Legacy)	Sends everything to a destination	Target Resource	None (Direct route)
Data Collection Rule (DCR)	What to collect, transform, and route	Azure Monitor	Needs a DCR-A to bind to resources
Data Collection Endpoint (DCE)	The network entry URI	Azure Monitor	Private Link / VNet Architecture
Azure Monitor Agent (AMA)	The local collection mechanism	Target VM / Arc Server	Depends entirely on a DCR

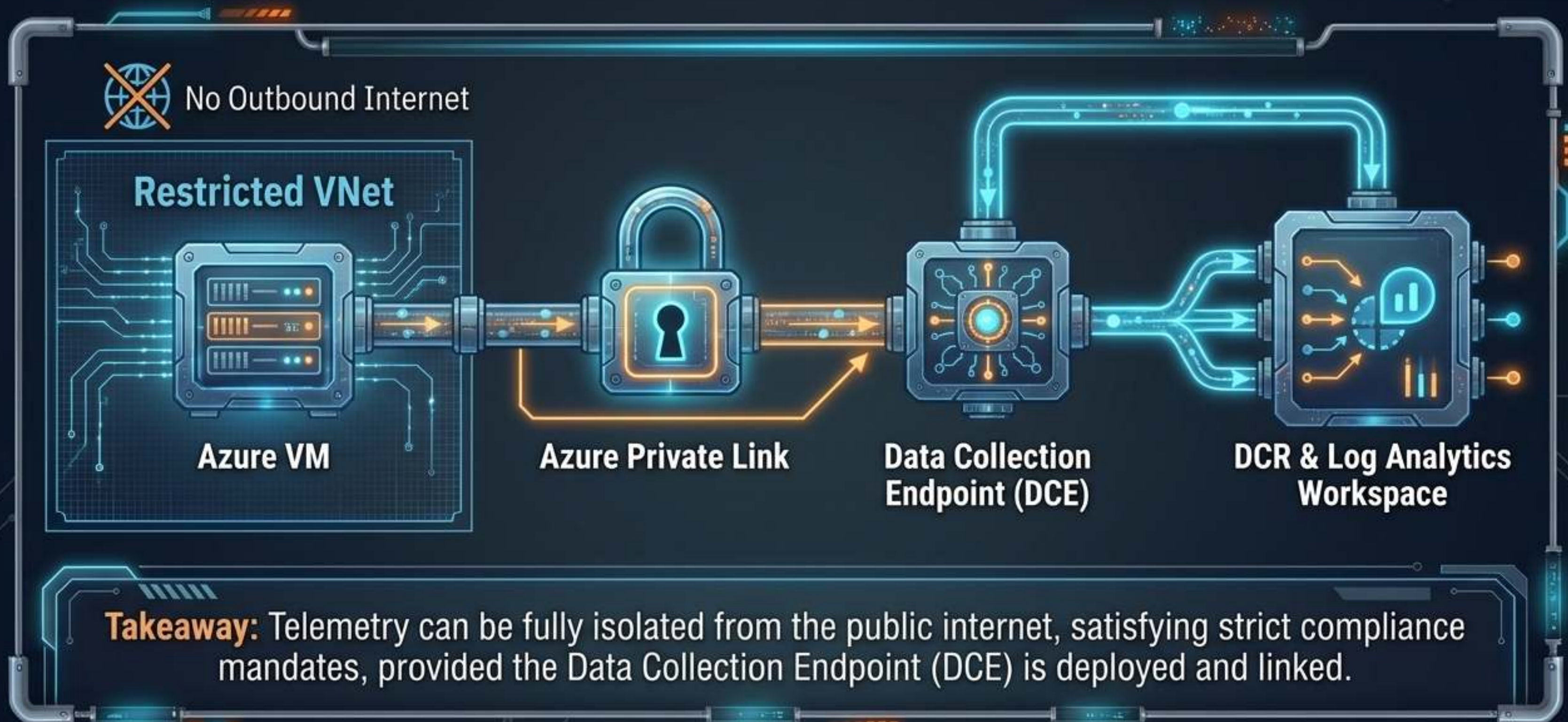
Scenario 1: Precision filtering for Windows Security Events



Scenario 2: Bridging on-premises telemetry via Azure Arc



Scenario 3: Locking down ingestion over private networks



Danger Zones: Common telemetry traps

The Filtering Trap

If an exam scenario requires filtering or redacting logs BEFORE they hit the workspace, the answer is always DCR, never Diagnostic Settings.

The Orphaned DCR Trap

Creating a DCR does absolutely nothing. It is an inert blueprint until an explicit Data Collection Rule Association (DCR-A) links it to a target resource.

The Deprecation Trap

The Microsoft Monitoring Agent (MMA) was officially retired in August 2024. Any architectural answer relying on MMA is now legacy; AMA is the sole correct path.

The Master Flow: From origin to intelligence

